



BGS

ENGINEERING

**BGS ENGINEERING
ALBUQUERQUE,
NEW MEXICO USA**

**THE MATERIAL THE MARKET
DOESN'T HAVE YET.**

**Every
structural
system is a
compromise.**

Metal is heavy, thermally limited, and has no intelligence

Conventional composites are light but passive — properties are added, not inherent

Every additional component adds weight, cost, complexity, and a new failure point

Nobody has solved the material itself

**The industry has
innovated around the
problem. Nobody has
solved it at the material
level.**

**These
properties are
intrinsic to the
material.
Not added.
Not bolted on.**

DARC is a proprietary compression-moldable carbon fiber composite with properties that do not exist together in any current production material.

Confirmed:

Lighter than aluminum and conventional carbon fiber prepreg

Exceptional impact toughness

Bulk electrical conductivity — memristive conditioning effect, floor not yet established

Through-thickness thermal conductivity

UV stable

Radio opaque

Complex monocoque geometry with molded-in hardware

Under active characterization:

Structural health monitoring

Triboelectric energy harvesting

EMI shielding

Radar signature characteristics

**DARC is
a material
platform. The
application list
grows with the
data.**

Near term:

**Drone and autonomous system frames —
monocoque, lightweight, tough, radar-relevant
Battery enclosures — grid storage, EV, electric
mobility**

Pipeline:

**Defense and aerospace structures
Industrial enclosures in UV-exposed or thermally
demanding environments
Any application where weight, toughness,
thermal performance, and structural intelligence
converge**

**High-rate
compression
molding. Proven
at bench scale.
Scaling to
production now.**

Press cycle approximately five minutes

Press decoupled from cure — molds transfer to ovens, press cycles immediately

Well over one hundred parts per press per shift

Each press and mold set added multiplies output linearly

Material cost exceptionally low relative to target sale price

Margins unusual in manufacturing, improving with volume

Three co-founders, equal partners, and an amazing ecosystem. Complementary by design.

A founding team whose collaboration spans 24 years
Started with a UV-resistant carbon fiber project,
brought it to Sandia National Laboratories, curiosity
took it somewhere much bigger

Daniel Brockett — Logistics & Operations

Former Formula D professional driver

US distributor for GKTech, an Australian automotive brand, across what may be
their largest single market

Matthew Gassner — Strategy & Analysis

Background in investments, banking, commodities trading, and data analytics

Identified the UV degradation problem, theorized the material solution, brought it to
Sandia through NMSBA

Gerald Stratton — Manufacturing & Mold Design

Precision machinist and mold maker, rare and diminishing skillset

Components for defense contractors, aerospace, and NASA Mars Cube One

Research & Validation Partners: Sandia National Laboratories (NMSBA) ·

NMSU wZ-Lab · NMSU PSL · Brad Jones, PhD · New Mexico Tech

New Mexico Ecosystem & Support: WESST · QStation · NM Startup Alliance

**· Arrowhead Center · APEX Accelerator · NMEDD · Loza & Loza · Southwest
Composite Works**

Eighteen months of hands-on development. Every property discovered, characterized, and validated by this team.

**The material
is validated.
The
process
works. The
network is
real.**

What comes next:

Production equipment installation and qualification

First prototype battery enclosure and drone frame

Full mechanical and electrical characterization data package

First contracts

Hiring in machining, quality control, and operations

Each press added is a multiplier